Assignment3 Report

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Dataset Introduction

This study uses the Australian Rental Market Data 2026 dataset obtained from Kaggle (Karunarathna 2026). The dataset provides a national snapshot of residential rental listings across Australia in 2026 and includes information on weekly rental prices, dwelling characteristics, and geographic attributes. To investigate rental price pressure across regions and dwelling types, variables related to rental price, location, dwelling category, and property characteristics were retained for analysis.

Before analysis, the dataset was reviewed for missing values, duplicate observations, and variable suitability. One duplicated observation was identified and removed to avoid repeated records. Missing values occurred in variables outside the scope of this study and therefore were not treated. Variables unrelated to the research objective, including listing text and detailed address information, were excluded to reduce unnecessary complexity. The final cleaned dataset retained variables describing rental price, geographic location, dwelling type, and property characteristics.

Dataset Description

The cleaned dataset contains 6766 observations and 9 variables. The retained variables include numeric measures describing rental price and dwelling characteristics (`price_display`, `bedrooms`, `bathrooms`, and `parking_spaces`) and categorical variables describing location and dwelling category (`state`, `locality`, `suburb`, and `propertyType`). These variables provide the basis for comparing rental conditions across Australia in 2026.

Dataset Summary

Table 1 summarises rental prices across dwelling types represented in the cleaned Australian rental dataset for 2026. Houses accounted for the largest share of observations (4,089 properties), followed by apartments (1,369 properties) and units (761 properties), indicating that the dataset primarily represents conventional residential dwellings. Median weekly rental prices differed across dwelling categories, ranging from AUD 430 for studios to AUD 970 for acreage and semi-rural properties.

Table 1: Summary statistics of rental prices by dwelling type.

propertyType	number_of_properties	avg_rent	median_rent
acreage/semi-rural	17	1094.1	970
terrace	7	871.4	750
apartment	1369	770.7	700
townhouse	142	741.3	700
house	4089	758.0	695
duplex/semi-detached	134	787.0	650
villa	25	624.8	590
unit	761	587.9	550

Table 1: Summary statistics of rental prices by dwelling type.

propertyType	number_of_properties	avg_rent	median_rent
other	33	525.0	540
flat	85	519.8	510
studio	104	484.2	430

In addition to dwelling type characteristics, Figure 1 shows the distribution of rental properties across Australia in 2026. Most observations were from New South Wales, followed by Victoria and Western Australia, indicating broad geographic coverage of the Australian rental market for subsequent analysis.

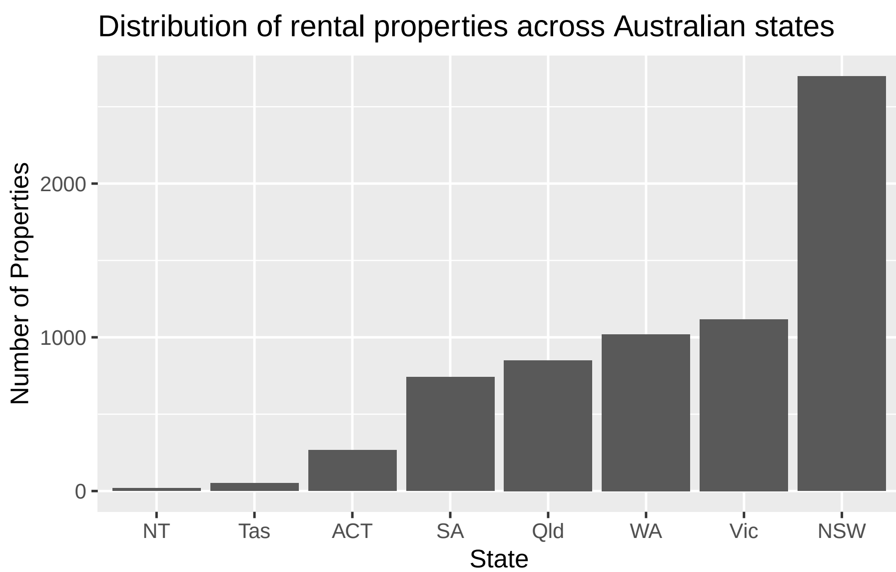

Figure 1: Distribution of rental properties across Australian states

Results

Discussion, Conclusion and Recommendations

References

Karunarathna, Kanchana. 2026. *Australian Rental Market Data 2026*. Kaggle. https://doi.org/10.34740/KAGGLE/DS/9582256.